

CVRN Study Notes

Basic science through advanced clinical practice. Every chapter runs the same route: the physiology first, then what changes in disease, then what it looks like at the bedside, then the guideline pathway stated explicitly, then questions where every option is explained, the wrong ones included.

The page opens as an outline. Nothing is hidden or locked, every section is one click away and Open everything is always available. Sections carry a time estimate so you can pick what fits the time you actually have.

Prepared by Dr. Sharilyn Rennie · Volume 1 · 50 minutes of reading · August 2026

CONTENTS

- | | | |
|----------|--|------------------------|
| 1 | Conduction and the Basic ECG
From pacemaker cell to rhythm strip | L1 14% · L2 8% |
| 2 | Coronary Artery Disease and Acute Coronary Syndrome
Oxygen supply, oxygen demand, and plaque rupture | L1 22% · L2 14% |
| 3 | Heart Failure
Neurohormonal compensation and its consequences | L1 20% · L2 13% |
| R | References
Textbooks and current clinical practice guidelines | 15 SOURCES |

CHAPTER 1 · L1 14% · L2 8% OF THE EXAM

Conduction and the Basic ECG

From pacemaker cell to rhythm strip

L1 14% · L2 8%

19 MIN

6 SECTIONS



Scan for the
Conduction and the
Basic ECG
question bank

Start with the physiology [PHYSIOLOGY]

Rhythm interpretation answers three questions, in this order:

- ① **Origin.** Where did the impulse begin?
- ② **Conduction.** What path did it take to the ventricles?
- ③ **Timing.** How long did each step take?

These three answers determine the rhythm. Naming it is the last step, not the first.

Automaticity. Pacemaker cells have no stable resting membrane potential. During phase 4, inward sodium and calcium current exceeds outward potassium current, so the membrane depolarises spontaneously until it reaches threshold and fires. This is the funny current. The cell group with the steepest phase 4 slope reaches threshold first and sets the heart rate, which under normal conditions is the sinoatrial node. ^{1,2}

INTRINSIC PACEMAKER RATES

SITE	INTRINSIC RATE (PER MINUTE)	ECG APPEARANCE IF IT TAKES OVER
Sinoatrial node	60 to 100	Upright P wave preceding every QRS in lead II
Atrial focus	60 to 80	Abnormally shaped or inverted P wave, narrow QRS
Atrioventricular junction	40 to 60	Narrow QRS, P wave absent, inverted, or following the QRS
Purkinje system or ventricular myocardium	20 to 40	Wide QRS, no associated P wave

The faster site suppresses the slower ones. A junctional or ventricular rhythm therefore means the sinoatrial node has failed, or conduction from it has been blocked. ^{1,7}

The conduction pathway, in sequence:

- ① The sinoatrial node depolarises.
- ② The impulse spreads across both atria, producing the **P wave**.

- ③ The atrioventricular node delays conduction. This delay allows atrial emptying to complete before ventricular systole, and it accounts for most of the **PR interval**.
- ④ The bundle of His, the right and left bundle branches, and the Purkinje fibres conduct rapidly through both ventricles, producing a narrow **QRS complex**.

Clinical consequence: an impulse that does not use this rapid pathway must travel myocyte to myocyte, which is slower. A QRS of 0.12 seconds or more therefore indicates abnormal ventricular activation, from bundle branch block, a ventricular focus, ventricular pacing, or a metabolic cause such as hyperkalaemia. ^{1,7,8}

What each part of the tracing represents [PATHOPHYSIOLOGY]

P wave. Atrial depolarisation. Upright in lead II indicates a sinus or high atrial origin conducting downward.

PR interval. Onset of P to onset of QRS. Normal 0.12 to 0.20 seconds, which is three to five small boxes.

QRS complex. Ventricular depolarisation. Normal under 0.12 seconds.

ST segment. The plateau phase of the ventricular action potential. Normally isoelectric with the baseline.

T wave. Ventricular repolarisation.

QT interval. Total duration of ventricular depolarisation and repolarisation. Shortens as rate increases, which is why it is rate corrected.

Atrial repolarisation occurs during the QRS complex and is not normally visible. ^{2,3}

MEMORY TOOL

Five questions, same order, every strip:

- ① Rate
- ② Regularity
- ③ P waves present and related to the QRS
- ④ PR interval
- ⑤ QRS width

Naming the rhythm before completing all five is the most common source of misinterpretation.

Measurement [PRACTICE]

Standard recording is 25 mm per second at a calibration of 10 mm per millivolt. One small box is 0.04 seconds, one large box is 0.20 seconds.

- ① **Regular rhythm.** Rate equals 300 divided by the number of large boxes between two R waves, or 1500 divided by the number of small boxes.
- ② **Irregular rhythm.** Neither shortcut applies. Count QRS complexes in a six second strip and multiply by ten.
- ③ **QT correction.** Bazett formula: QTc equals QT divided by the square root of the R to R interval in seconds. A QTc above approximately 500 ms is associated with markedly increased risk of torsades de pointes.

Measure from the earliest onset to the latest offset across several leads, since a single lead may not show the full duration. ^{7,9}

Atrioventricular block [PATHOPHYSIOLOGY]

All three degrees answer two questions: does every atrial impulse reach the ventricles, and is the PR interval constant?

- ① **First degree.** PR interval greater than 0.20 seconds, constant, with every P wave conducted. A conduction delay rather than a true block.
- ② **Second degree, Mobitz type I (Wenckebach).** PR interval lengthens progressively until one P wave is not conducted, then the sequence resets. The lesion is usually within the atrioventricular node. Often responds to atropine.
- ③ **Second degree, Mobitz type II.** PR interval constant, then a P wave fails to conduct without warning. The lesion is usually infranodal. Carries a risk of progression to complete block, and the escape rhythm below the lesion is slow and unreliable.
- ④ **Third degree (complete).** No atrial impulses conduct. P waves and QRS complexes occur at independent rates with no fixed relationship. This atrioventricular dissociation is the diagnostic finding.

Because the lesion in Mobitz type II and complete block usually sits below the atrioventricular node, atropine frequently fails, and temporary pacing capability is prepared at the same time it is given. ^{8,5}

MEMORY TOOL

Longer, longer, longer, drop, then you have a Wenckebach. Constant PR with a dropped beat is Mobitz II. P waves and QRS complexes marching independently is third degree. The distinguishing step in all three is measuring the PR interval across several consecutive beats rather than looking at one complex.

AT THE BEDSIDE

Treatment follows symptoms, not the number. A rate of 40 in a well perfused, asymptomatic patient is a finding. A rate of 52 with hypotension, altered mental status, ischaemic chest discomfort, or acute heart failure meets the definition of symptomatic bradycardia and is treated.

Sequence for symptomatic bradycardia: atropine 1 mg intravenous, repeated every three to five minutes to a maximum of 3 mg, while preparing transcutaneous pacing. If atropine is ineffective, proceed to transcutaneous pacing or a dopamine or epinephrine infusion. ⁸

Test yourself [QUESTIONS]

A monitored patient has a ventricular rate of 44. The PR interval measures 0.16 seconds and is identical on every conducted beat, and every third P wave is not followed by a QRS complex. The patient is diaphoretic with a blood pressure of 78 over 44. What is the priority action?

A Continue observation and reassess the rhythm in 15 minutes

×

Observation is appropriate only when perfusion is adequate. Diaphoresis and a systolic pressure of 78 meet the criteria for symptomatic bradycardia, which requires treatment.

B Administer atropine and prepare for transcutaneous pacing at the same time ✓ correct

Correct. A constant PR interval with intermittently non-conducted P waves is Mobitz type II second degree block, which is usually infranodal. Atropine acts on the atrioventricular node and often fails when the lesion is below it, so pacing capability is prepared concurrently rather than after atropine fails.

C Administer adenosine to slow conduction and expose the underlying rhythm

×

Adenosine blocks conduction through the atrioventricular node. In a patient already failing to conduct through that node and who is hypotensive, this risks prolonged asystole. Adenosine is indicated for regular narrow complex tachycardia.

D Administer a 500 mL isotonic fluid bolus and reassess

×

The hypotension here results from an inadequate ventricular rate rather than from hypovolaemia. Fluid will not raise a rate of 44, and giving it first delays the indicated treatment.

WHY THIS MATTERS

Two things are tested together: identifying Mobitz type II by a constant PR interval with dropped beats, and knowing that the anatomic level of the block determines

whether atropine can be relied upon. Recognising the rhythm but managing it as Mobitz type I still produces the wrong action.

MEMORY TOOL

Constant PR with a dropped beat is **M**obitz II, so **M**ake ready to pace.

A rhythm strip shows a QRS duration of 0.16 seconds, no identifiable P waves, and a regular rate of 38. Which conclusion is best supported?

A The impulse originates in the sinoatrial node with delayed conduction ✕

A sinus origin produces visible P waves. Their complete absence excludes it.

B The impulse originates in the atrioventricular junction ✕

A junctional focus lies above the ventricles and conducts through the intact His-Purkinje system, producing a narrow QRS at 40 to 60 per minute. The QRS duration of 0.16 seconds excludes this.

C The impulse originates below the bundle branches, in the Purkinje system or ventricular myocardium ✓ correct

Correct. A QRS of 0.16 seconds indicates activation outside the rapid conduction system, and a rate of 38 falls within the intrinsic ventricular escape range of 20 to 40. This is an escape rhythm maintaining cardiac output, so suppressing it with an antiarrhythmic would remove the only functioning pacemaker.

D This is atrial flutter with variable atrioventricular conduction ✕

Atrial flutter produces organised atrial activity at approximately 250 to 350 per minute with a ventricular rate that is a fraction of it. Neither finding is present.

WHY THIS MATTERS

QRS width localises the origin and rate identifies which escape pacemaker is active. Reading them together locates the focus without memorising rhythm names, and it establishes the management principle: an escape rhythm is supported, never suppressed.

MEMORY TOOL

Wide means below. A narrow QRS used the conduction system, so the origin is at or above the junction. A wide QRS did not.

Sources for this chapter [PRACTICE]

1. Hall JE, Hall ME. **Guyton and Hall Textbook of Medical Physiology**. 14th ed. Elsevier; 2021.
2. Tortora GJ, Derrickson B. **Principles of Anatomy and Physiology**. 16th ed. Wiley; 2020.
3. Marieb EN, Hoehn K. **Human Anatomy & Physiology**. 12th ed. Pearson; 2022.
5. Urden LD, Stacy KM, Lough ME. **Critical Care Nursing: Diagnosis and Management**. 9th ed. Elsevier; 2022.
7. Wagner GS, Strauss DG. **Marriott's Practical Electrocardiography**. 13th ed. Wolters Kluwer; 2021.
8. Kusumoto FM, Schoenfeld MH, Barrett C, et al. 2018 ACC/AHA/HRS Guideline on the Evaluation and Management of Patients With Bradycardia and Cardiac Conduction Delay. **Circulation**. 2019.
9. Sandau KE, Funk M, Auerbach A, et al. Update to Practice Standards for Electrocardiographic Monitoring in Hospital Settings: A Scientific Statement From the American Heart Association. **Circulation**. 2017.

Coronary Artery Disease and Acute Coronary Syndrome

Oxygen supply, oxygen demand, and plaque rupture

L1 22% • L2 14%

14 MIN

6 SECTIONS



Scan for the
Coronary Artery
Disease and Acute
Coronary Syndrome
question bank

Start with the physiology [PHYSIOLOGY]

Myocardium is obligately aerobic and extracts approximately 70 to 80 percent of delivered oxygen at rest, compared with about 25 percent in most other tissues. Because extraction is already near maximal, the myocardium cannot meet increased demand by extracting more oxygen. It must increase coronary blood flow. ^{1,4}

Coronary perfusion occurs during diastole. Systolic contraction compresses the intramyocardial vessels, particularly in the left ventricular subendocardium. Two clinical consequences follow directly:

- ① **Tachycardia reduces perfusion.** Increasing heart rate shortens diastole proportionally more than systole, so perfusion time falls at the moment demand is highest.
- ② **The subendocardium is affected first.** It is the last layer perfused and the most vulnerable to ischaemia, which is why subendocardial ischaemia produces ST depression rather than ST elevation.

MEMORY TOOL

Myocardial oxygen demand has three determinants: **heart rate, contractility, and wall tension** (preload and afterload). Every antianginal agent reduces at least one of them. Beta blockers reduce rate and contractility. Nitrates reduce preload. Afterload reducers reduce wall tension. Identifying which determinant a drug acts on gives you its indication without memorising a list. ⁴

From stable plaque to acute event [PATHOPHYSIOLOGY]

Stable angina. A fixed stenosis limits flow reserve. Perfusion is adequate at rest and inadequate on exertion, producing predictable, reproducible, exertional chest discomfort relieved by rest or nitroglycerin. ¹²

Acute coronary syndrome. Plaque rupture or erosion exposes thrombogenic material to circulating blood. Platelets adhere and aggregate and a thrombus forms. What follows depends on whether the thrombus occludes the vessel completely. ^{10,11}

THE ACUTE CORONARY SYNDROME SPECTRUM

	UNSTABLE ANGINA	NSTEMI	STEMI
Occlusion	Partial	Partial or transient	Complete
ECG	Normal, ST depression, or T wave inversion	Normal, ST depression, or T wave inversion	ST elevation, or new left bundle branch block
Troponin	Not elevated	Elevated	Elevated
Necrosis	None	Subendocardial	Transmural
Priority	Antithrombotic therapy, risk stratification	Antithrombotic therapy, early invasive strategy	Immediate reperfusion

Unstable angina and NSTEMI are distinguished only by troponin. They can present identically, which is why serial biomarkers are obtained rather than the diagnosis being made on presentation alone. ^{10,11}

Biomarkers [PATHOPHYSIOLOGY]

High sensitivity cardiac troponin detects very small quantities of myocardial injury. It has high sensitivity and lower specificity, so it excludes myocardial infarction well and confirms it less well. Troponin elevation occurs in sepsis, pulmonary embolism, myocarditis, renal impairment, tachyarrhythmia, and decompensated heart failure, none of which are coronary occlusion. ¹¹

- ① **Myocardial injury** is any troponin elevation above the 99th percentile upper reference limit.
- ② **Acute injury** requires a rise or fall across serial measurements, the delta.
- ③ **Myocardial infarction** requires acute injury plus clinical evidence of ischaemia: symptoms, ECG changes, imaging evidence, or angiographic findings.

A stably elevated troponin in chronic kidney disease is chronic myocardial injury. A value that rises significantly over one to three hours is acute. ¹¹

THE GUIDELINE PATHWAY

Reperfusion timing for STEMI, per the 2025 ACC/AHA acute coronary syndrome guideline: ¹⁰

- ① At a PCI capable facility, target first medical contact to device within **90 minutes**.
- ② Requiring transfer, target first medical contact to device within **120 minutes**.
- ③ If 120 minutes cannot be met, give fibrinolytic therapy within **30 minutes** of arrival provided there is no contraindication, then transfer.

The decision rests on anticipated time to reperfusion, not on institutional preference.

Absolute contraindications to fibrinolysis: any prior intracranial haemorrhage, known structural cerebral vascular lesion or malignant intracranial neoplasm, ischaemic stroke within three months, suspected aortic dissection, active bleeding or bleeding diathesis, and significant closed head or facial trauma within three months. ¹⁰

Mechanical complications after infarction [PATHOPHYSIOLOGY]

These typically occur between day two and day seven after a large infarct and present as abrupt decompensation in a previously stable patient. ^{4,5}

Papillary muscle rupture. Acute severe mitral regurgitation. Sudden dyspnoea, new holosystolic murmur, pulmonary oedema, hypotension.

Ventricular septal rupture. New harsh holosystolic murmur with a palpable thrill, biventricular failure, oxygen saturation step up between right atrium and right ventricle.

Free wall rupture. Tamponade producing pulseless electrical activity.

Cardiogenic shock. Most often after a large anterior infarct. Low cardiac output with elevated filling pressures.

MEMORY TOOL

A murmur that was not present the day before, in a patient who infarcted two to seven days ago, is a mechanical complication until proven otherwise.

Test yourself [QUESTIONS]

A patient with a large anterior STEMI is on hospital day three. He develops sudden severe dyspnoea, a new loud holosystolic murmur at the apex, and a blood pressure of 82 over 50. Which explanation best fits?

A Reinfarction of the anterior wall

✗

Reinfarction presents with recurrent ischaemic chest discomfort and new ECG changes. It does not produce a new holosystolic murmur, which is the finding that directs the diagnosis elsewhere.

B Papillary muscle rupture producing acute mitral regurgitation

✓ correct

Correct. The timing, day two to seven after infarction, and the combination of a new holosystolic murmur with abrupt haemodynamic decompensation are characteristic. The left atrium has not dilated or become compliant, so the regurgitant volume transmits directly to the pulmonary circulation, producing pulmonary oedema and shock rather than gradual decline.

C Acute pericarditis

✗

Post-infarction pericarditis occurs in this window but produces a pericardial friction rub and positional pleuritic chest pain, not a holosystolic murmur with hypotension and pulmonary oedema.

D Anxiety related hyperventilation

✗

This attributes objective findings, a new murmur and a systolic pressure of 82, to a subjective cause. Hyperventilation produces neither.

WHY THIS MATTERS

Timing plus a new murmur plus haemodynamic collapse is the pattern. Acute and chronic mitral regurgitation behave differently because of atrial compliance: the

chronically volume loaded atrium has remodelled and accommodates the regurgitant volume at lower pressure, while the acutely affected atrium has not.

MEMORY TOOL

Day two to seven, new murmur, sudden decompensation. Consider a mechanical complication before a medical one.

Sources for this chapter [PRACTICE]

1. Hall JE, Hall ME. **Guyton and Hall Textbook of Medical Physiology**. 14th ed. Elsevier; 2021.
4. Libby P, Bonow RO, Mann DL, et al, eds. **Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine**. 12th ed. Elsevier; 2022.
5. Urden LD, Stacy KM, Lough ME. **Critical Care Nursing: Diagnosis and Management**. 9th ed. Elsevier; 2022.
10. Rao SV, O'Donoghue ML, Ruel M, et al. 2025 ACC/AHA/ACEP/NAEMSP/SCAI Guideline for the Management of Patients With Acute Coronary Syndromes. **Circulation**. 2025.
11. Thygesen K, Alpert JS, Jaffe AS, et al. Fourth Universal Definition of Myocardial Infarction (2018). **Circulation**. 2018.
12. Virani SS, Newby LK, Arnold SV, et al. 2023 AHA/ACC/ACCP/ASPC/NLA/PCNA Guideline for the Management of Patients With Chronic Coronary Disease. **Circulation**. 2023.

Heart Failure

Neurohormonal compensation and its consequences

L1 20% · L2 13%

17 MIN

7 SECTIONS



Scan for the
Heart Failure
question bank

Start with the physiology [PHYSIOLOGY]

Cardiac output equals stroke volume multiplied by heart rate. Stroke volume is determined by three factors:

- ① **Preload.** Ventricular end-diastolic volume, the degree of myocardial fibre stretch before contraction.
- ② **Afterload.** The resistance the ventricle must overcome to eject.
- ③ **Contractility.** The intrinsic force of contraction, independent of loading conditions.

Heart failure is the clinical syndrome in which cardiac structural or functional impairment prevents the ventricle from delivering an output adequate to metabolic demand, or does so only at elevated filling pressures.¹³ Both presentations follow from that definition: reduced forward output, and congestion from elevated filling pressures. Most patients have some of each.

The Frank-Starling relationship. Increasing sarcomere stretch improves actin and myosin overlap and increases the force of the subsequent contraction, up to an optimum. In the failing ventricle the curve is flattened and shifted rightward, so a given increase in filling volume produces less additional stroke volume and a substantially greater rise in filling pressure. This is why volume loading a failing ventricle produces congestion rather than improved output.^{1,4}

Compensatory mechanisms [PRACTICE]

When cardiac output falls, the body activates the same reflexes it uses for hypovolaemia. Each is beneficial acutely and harmful when sustained.^{4,13}

- ① **Sympathetic activation.** Increases heart rate and contractility and causes arteriolar vasoconstriction. Sustained, it increases myocardial oxygen demand, shortens diastolic filling and coronary perfusion time, raises afterload, and is directly cytotoxic to myocytes.
- ② **Renin angiotensin aldosterone activation.** Reduced renal perfusion triggers renin release. Angiotensin II causes vasoconstriction and aldosterone causes sodium and water

retention. Sustained, it adds both preload and afterload and promotes myocardial fibrosis.

- ③ **Ventricular remodelling.** Under sustained neurohormonal stimulation the ventricle dilates and hypertrophies into a geometry that is mechanically less efficient, converting an initial injury into a progressive disease.

MEMORY TOOL

Each of the four pillars of therapy **opposes** one of these compensations. That is why a beta blocker is given to a failing heart and an afterload reducer to a patient whose blood pressure is already low. The target is the neurohormonal cascade driving remodelling, not the pump itself. ¹³

The four pillars [PRACTICE]

GUIDELINE DIRECTED MEDICAL THERAPY FOR HEART FAILURE WITH REDUCED EJECTION FRACTION

PILLAR	MECHANISM OPPOSED	MONITORING
ARNI, or ACE inhibitor or ARB	Renin angiotensin aldosterone system; ARNI additionally preserves natriuretic peptides	Potassium, creatinine, blood pressure, angioedema. A 36 hour washout is required when switching from an ACE inhibitor to an ARNI.
Beta blocker	Sympathetic activation and remodelling	Heart rate, blood pressure. Initiate at low dose and titrate slowly. Do not initiate during acute decompensation.
Mineralocorticoid receptor antagonist	Aldosterone mediated sodium retention and fibrosis	Potassium and renal function. The pillar most often responsible for hyperkalaemia.
SGLT2 inhibitor	Multiple, including natriuresis and altered myocardial substrate use	Volume status, genital mycotic infection, euglycaemic ketoacidosis. Benefit is independent of diabetes status.

A modest rise in creatinine after initiating an ACE inhibitor, ARNI, or mineralocorticoid receptor antagonist reflects expected haemodynamic change rather than renal injury, and is not by itself an indication to discontinue therapy. ¹³

Assessing congestion and perfusion [PRACTICE]

Two independent questions determine management. The resulting four profiles were described by Stevenson and validated by Nohria and colleagues. ¹⁴

- 1 **Is the patient congested?** Wet or dry. Assessed by jugular venous pressure, orthopnoea, crackles, peripheral oedema, and hepatomegaly.
- 2 **Is the patient adequately perfused?** Warm or cold. Assessed by pulse pressure, extremity temperature, mental status, and renal function.

HAEMODYNAMIC PROFILES AND MANAGEMENT PRIORITY

	DRY (NO CONGESTION)	WET (CONGESTED)
Warm (perfused)	Profile A. Compensated. Continue therapy.	Profile B. Diuresis.
Cold (hypoperfused)	Profile L. Underfilled or over-diuresed. Cautious volume.	Profile C. Highest mortality. Decongestion with consideration of inotropic support.

Profile C is the most frequently mismanaged. Crackles suggest aggressive diuresis, but low output means aggressive diuresis alone may further reduce perfusion. A narrow pulse pressure, cool extremities, altered mental status, and rising creatinine identify the cold patient. ^{14,5}

Self care and transition of care [PRACTICE]

Readmission within 30 days is driven substantially by what occurs in the first two weeks after discharge. Effective teaching is specific and gives the patient an action threshold rather than general advice. ^{13,6}

- 1 **Daily weight.** On waking, after voiding, in similar clothing, on the same scale. Consistency matters because the trend is the signal.
- 2 **Call threshold.** A gain of two to three pounds in 24 hours, or five pounds in one week.
- 3 **Sodium.** Guidance the patient can act on: read labels, avoid processed and restaurant foods, which contribute most dietary sodium.
- 4 **Symptom escalation.** Distinguish what warrants a call today from what warrants emergency evaluation.

Test yourself [QUESTIONS]

A patient admitted with decompensated heart failure has crackles to the mid lung fields, jugular venous distension, cool extremities, a narrow pulse pressure, and a creatinine that has risen from 1.1 to 1.7 mg/dL. Which management approach is most appropriate?

A Aggressive intravenous diuresis as the sole intervention

✗

This addresses congestion and ignores perfusion. Cool extremities, narrow pulse pressure, and rising creatinine indicate hypoperfusion. Diuresis alone in profile C can further reduce cardiac output and worsen renal function.

B Decongestion with consideration of inotropic support

✓ correct

Correct. Congestion with hypoperfusion is profile C, which carries the highest mortality of the four profiles. Congestion must be treated, but perfusion requires simultaneous support, which is why inotropic therapy is considered here and not in the warm and wet patient.

C Intravenous fluid resuscitation to improve renal perfusion

✗

The rising creatinine suggests this, but crackles and jugular venous distension indicate volume overload. The renal impairment reflects low forward output and venous congestion, not hypovolaemia. Volume would worsen both.

D Withhold all guideline directed therapy and observe

✗

Temporarily withholding beta blockade may be appropriate in low output states, but withholding all therapy while the patient is both congested and hypoperfused is not a management plan.

WHY THIS MATTERS

The profile framework exists to prevent the first option. Congestion and perfusion are assessed independently and both answers are required before a plan is made. Renal

function that worsens during decongestion often reflects venous congestion rather than volume depletion, which is the opposite of what the creatinine alone suggests.

MEMORY TOOL

Wet means decongest. Cold means support forward flow. A cold and wet patient needs both, not one.

Sources for this chapter [PRACTICE]

1. Hall JE, Hall ME. **Guyton and Hall Textbook of Medical Physiology**. 14th ed. Elsevier; 2021.
4. Libby P, Bonow RO, Mann DL, et al, eds. **Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine**. 12th ed. Elsevier; 2022.
5. Urden LD, Stacy KM, Lough ME. **Critical Care Nursing: Diagnosis and Management**. 9th ed. Elsevier; 2022.
6. Sole ML, Klein DG, Moseley MJ. **Introduction to Critical Care Nursing**. 8th ed. Elsevier; 2021.
13. Heidenreich PA, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA Guideline for the Management of Heart Failure. **Circulation**. 2022.
14. Nohria A, Tsang SW, Fang JC, et al. Clinical assessment identifies hemodynamic profiles that predict outcomes in patients admitted with heart failure. **J Am Coll Cardiol**. 2003.

References

Physiology and anatomy from current standard texts; clinical management from the most recent published guideline in each area. Guideline year is stated so a superseded version is obvious at a glance.

1. Hall JE, Hall ME. **Guyton and Hall Textbook of Medical Physiology**. 14th ed. Elsevier; 2021.

2. Tortora GJ, Derrickson B. **Principles of Anatomy and Physiology**. 16th ed. Wiley; 2020.

3. Marieb EN, Hoehn K. **Human Anatomy & Physiology**. 12th ed. Pearson; 2022.

4. Libby P, Bonow RO, Mann DL, et al, eds. **Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine**. 12th ed. Elsevier; 2022.

5. Urden LD, Stacy KM, Lough ME. **Critical Care Nursing: Diagnosis and Management**. 9th ed. Elsevier; 2022.

6. Sole ML, Klein DG, Moseley MJ. **Introduction to Critical Care Nursing**. 8th ed. Elsevier; 2021.

7. Wagner GS, Strauss DG. **Marriott's Practical Electrocardiography**. 13th ed. Wolters Kluwer; 2021.

8. Kusumoto FM, Schoenfeld MH, Barrett C, et al. 2018 ACC/AHA/HRS Guideline on the Evaluation and Management of Patients With Bradycardia and Cardiac Conduction Delay. **Circulation**. 2019.

9. Sandau KE, Funk M, Auerbach A, et al. Update to Practice Standards for Electrocardiographic Monitoring in Hospital Settings: A Scientific Statement From the American Heart Association. **Circulation**. 2017.

10. Rao SV, O'Donoghue ML, Ruel M, et al. 2025 ACC/AHA/ACEP/NAEMSP/SCAI Guideline for the Management of Patients With Acute Coronary Syndromes. **Circulation**. 2025.
<https://www.ahajournals.org/doi/10.1161/CIR.0000000000001309>

11. Thygesen K, Alpert JS, Jaffe AS, et al. Fourth Universal Definition of Myocardial Infarction (2018). **Circulation**. 2018.

12. Virani SS, Newby LK, Arnold SV, et al. 2023 AHA/ACC/ACCP/ASPC/NLA/PCNA Guideline for the Management of Patients With Chronic Coronary Disease. **Circulation**. 2023.

13. Heidenreich PA, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA Guideline for the Management of Heart Failure. **Circulation**. 2022.
<https://www.ahajournals.org/doi/10.1161/CIR.0000000000001063>

14. Nohria A, Tsang SW, Fang JC, et al. Clinical assessment identifies hemodynamic profiles that predict outcomes in patients admitted with heart failure. **J Am Coll Cardiol**. 2003.

15. Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults. **Hypertension**. 2018.

CVRN Study Notes, Volume 1. Prepared by Dr. Sharilyn Rennie for the ECG & CVRN Review Course, MedMasters Collaborative. Teaching material for certification preparation, not a clinical protocol.